TUTORIAL 3

Linear Systems and Subspaces

MOTIVATION AND GUIDING IDEA

Row reduction replaces a problem with an equivalent one that is easier to read. After calculating, return to the question: is the system consistent, which variables are free, and what is the solution set? For a nonhomogeneous system, the difference of two solutions solves the associated homogeneous system. This explains the form ``a particular solution plus a homogeneous solution.''

NOTATION AND CONVENTIONS

If $A \in M_{m\times n}(F)$, then $Ax = b$ is a system of m equations in n variables, with $x \in F^n$ and $b \in F^m$. The notation $[A \mid b]$ denotes its augmented matrix, and $A \sim B$ denotes row equivalence. The rank $\operatorname{rank} A$ is the number of pivots. Define $\ker A = \{x \in F^n : Ax = 0\}$. The expression $\operatorname{span}\{v_1, \ldots, v_k\}$ denotes all combinations $\alpha_1 v_1 + \cdots + \alpha_k v_k$ with $\alpha_j \in F$. The notation $x_0 + U = \{x_0 + u : u \in U\}$ denotes a translate of a subspace U. Distinguish the empty set $\varnothing$, with no elements, from $\{0\}$, which has one element.

DEFINITION 3.1

Vector space axioms and solution tools. A vector space over a field F is a set V equipped with operations $V \times V \to V$ and $F \times V \to V$ satisfying, for all $u, v, w \in V$ and $\alpha, \beta \in F$,

$$u + v = v + u, \qquad (u + v) + w = u + (v + w).$$

There is $0_V \in V$ such that $v + 0_V = v$, and every $v \in V$ has an additive inverse $-v \in V$ satisfying $v + (-v) = 0_V$. In addition,

$$\alpha(u + v) = \alpha u + \alpha v, \qquad (\alpha + \beta)v = \alpha v + \beta v,$$

$$(\alpha\beta)v = \alpha(\beta v), \qquad 1_F v = v.$$

For $A \in M_{m\times n}(F)$, the system $Ax = b$ is consistent if and only if $\operatorname{rank} A = \operatorname{rank}[A \mid b]$. If x_0 is a particular solution, then its full solution set is $x_0 + \ker A$: indeed, $Ax = b$ is equivalent to $A(x - x_0) = 0$. This solution set is a subspace of F^n if and only if $b = 0$, since containing the zero vector forces $A0 = b = 0$.

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MOTIVATION AND GUIDING IDEA

Objectives. We will formulate problems as linear systems, perform row reduction while preserving the solution set, and distinguish free variables from pivot variables. We will connect homogeneous systems with vector spaces and test closure systematically.

3.1 Linear systems and row reduction

3.1.1 A system of linear equations

RECALL AND TOOLS

In general, for a positive integer n, a linear equation in n unknowns (variables) has the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

Here $a_1, a_2, ..., a_n$ and b are **given scalars** in a specified field $\mathbb{F}$, such as $\mathbb{R}$, $\mathbb{C}$, or a finite field, and $x_1, x_2, ..., x_n$ are the **unknowns**, or variables.
The scalar a_i is the **coefficient** of x_i, and b is the **constant term** of the equation.
A system of m equations in n unknowns has the general form

$$\begin{cases} a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,n}x_n = b_1 \\ a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,n}x_n = b_2 \\ \vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n}x_n = b_m \end{cases}$$

To solve the system, it is convenient to collect the **coefficients** in a matrix, preserving the variable order. To shorten the notation, we write either the matrix of variable coefficients, called the **coefficient matrix**, or the **augmented matrix**, which also includes the constant terms. Following the basic rules consistently makes it clear which coefficient belongs to each variable.

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EXAMPLE 3.1

$$\begin{cases} 2x_1 - x_2 = 0 \\ -4x_1 + 3x_2 = 1 \end{cases} \xrightarrow{\text{matrix form}} \left(\begin{array}{cc|c} 2 & -1 & 0 \\ -4 & 3 & 1 \end{array}\right)$$

3.1.2 Elementary operations

DEFINITION 3.2

Elementary row operations:
An elementary row operation modifies a matrix in one of the following ways:

1. Interchange rows i and j, denoted by $R_i \longleftrightarrow R_j$.
2. Multiply row i by a **nonzero** scalar λ, denoted by $R_i \longleftarrow \lambda R_i$.
   (Dividing by a nonzero scalar a is multiplication by $1/a$.)

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3. Add λ times row j to a different row i, where $i \neq j$, denoted by $R_i \longleftarrow R_i + \lambda R_j$.

- Elementary row operations preserve the solutions of a linear system.
- Every elementary row operation has an inverse that is also an elementary row operation.

DEFINITION 3.3

Matrices A and B are **row equivalent** (or simply equivalent), denoted by $A \sim B$, if B is obtained by applying a finite sequence of elementary row operations to A.

DEFINITION 3.4

Leading entry (pivot): in every nonzero row of a matrix, the first nonzero entry is called its leading entry.

DEFINITION 3.5

Row echelon form: each leading entry is strictly to the left of the leading entry in the next nonzero row. Any zero rows appear below all nonzero rows.

DEFINITION 3.6

Reduced row echelon form: a matrix in row echelon form such that, in each nonzero row, the leading entry equals 1, and every other entry above and below it in the same column equals 0. These requirements apply to every column, including the constant column of an augmented matrix.

PROPOSITION 3.1

Every matrix is row equivalent to a unique matrix in reduced row echelon form.

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Examples:

1. $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ is not in row echelon form.
2. $\begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ is in row echelon form, but not in reduced row echelon form.
3. $\begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \end{pmatrix}$ is in reduced row echelon form.

3.1.3 Gaussian elimination

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RECALL AND TOOLS

1. Start with the first active row and the first column. Look in that column for a nonzero entry in the active row or below it. If none exists, move to the next column.
2. Interchange rows to bring this entry into the active row. Division by it is permitted because it has been checked to be nonzero.
3. Multiply the active row by the inverse of its leading entry to obtain a leading 1, and use it to eliminate the entries below.
4. Move to the next active row and seek a pivot column to the right of the previous one. A pivot column need not have the same index as its row.
5. After reaching row echelon form, eliminate the entries above each pivot, working from the last pivot to the first, to obtain reduced row echelon form. This completes Gauss-Jordan elimination.

EXERCISE 3.1

Over $\mathbb{R}$, reduce the augmented matrix

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 2 & 1 & 3 & 12 \\ 1 & 2 & 2 & 6 \end{array}\right)$$

to reduced row echelon form, and read off the solution of the system.

WORKED SOLUTION

Subtract multiples of the first row:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 2 & 1 & 3 & 12 \\ 1 & 2 & 2 & 6 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & -1 & 1 & 4 \\ 0 & 1 & 1 & 2 \end{array}\right).$$

Multiply R_2 by -1, and subtract it from R_3:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & -1 & -4 \\ 0 & 0 & 2 & 6 \end{array}\right) \xrightarrow{R_3 \leftarrow R_3/2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & -1 & -4 \\ 0 & 0 & 1 & 3 \end{array}\right).$$

Apply $R_2 \leftarrow R_2 + R_3$, followed by $R_1 \leftarrow R_1 - R_3 - R_2$:

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right).$$

The unique solution is therefore $(2, -1, 3)^T$. Substitution gives $2 - 1 + 3 = 4$, $4 - 1 + 9 = 12$, and $2 - 2 + 6 = 6$, as required.

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NOTE

To solve a linear system, reduce its **augmented matrix**, including the constant terms. You may stop at row echelon form and use back substitution, or continue to reduced row echelon form and read off the solutions.

EXAMPLE 3.2

Consider the following system:

$$\begin{cases} 2x + 2y + 2z = 8 \\ 4x + 2z = 12 \\ 2x + 2y = 8 \end{cases}$$

Write the system in matrix form:

$$\left(\begin{array}{ccc|c} 2 & 2 & 2 & 8 \\ 4 & 0 & 2 & 12 \\ 2 & 2 & 0 & 8 \end{array}\right)$$

Applying the same row-reduction method gives

$$\left(\begin{array}{ccc|c} 2 & 2 & 2 & 8 \\ 4 & 0 & 2 & 12 \\ 2 & 2 & 0 & 8 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

Returning to the equations,

$$\begin{cases} 1 \cdot x + 0 \cdot y + 0 \cdot z = 3 \\ 0 \cdot x + 1 \cdot y + 0 \cdot z = 1 \\ 0 \cdot x + 0 \cdot y + 1 \cdot z = 0 \end{cases} \longrightarrow \begin{cases} x = 3 \\ y = 1 \\ z = 0 \end{cases}$$

we find that the solution set of the system is $S = \{\begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}\}$.

This example shows how row reduction helps solve a linear system. We will next see that a system need not have a unique solution.

EXERCISE 3.2

Reduce the following matrix over $\mathbb{Z}_7 = \mathbb{Z}/7\mathbb{Z}$ to reduced row echelon form. Please perform all operations modulo 7, using the representatives $0, 1, \ldots, 6$:

$$\left(\begin{array}{cccc|c} 1 & 0 & 5 & 0 & 1 \\ 0 & 1 & 2 & 1 & 2 \\ 3 & 4 & 0 & 1 & 3 \end{array}\right)$$

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EXERCISE 3.2 CONTINUED

WORKED SOLUTION

$$\left(\begin{array}{cccc|c} 1 & 0 & 5 & 0 & 1 \\ 0 & 1 & 2 & 1 & 2 \\ 3 & 4 & 0 & 1 & 3 \end{array}\right) \xrightarrow{R_3 \to R_3 - 3R_1} \left(\begin{array}{cccc|c} 1 & 0 & 5 & 0 & 1 \\ 0 & 1 & 2 & 1 & 2 \\ 0 & 4 & 6 & 1 & 0 \end{array}\right) \xrightarrow{R_3 \to R_3 - 4R_2}$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 5 & 0 & 1 \\ 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & 5 & 4 & 6 \end{array}\right)$$

$$\xrightarrow{R_3 \to 5^{-1}R_3} \left(\begin{array}{cccc|c} 1 & 0 & 5 & 0 & 1 \\ 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & 1 & 5 & 4 \end{array}\right)$$

$$\xrightarrow[R_2 \to R_2 - 2R_3]{R_1 \to R_1 - 5R_3} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 3 & 2 \\ 0 & 1 & 0 & 5 & 1 \\ 0 & 0 & 1 & 5 & 4 \end{array}\right)$$

EXERCISE 3.3

Please solve the following system over $\mathbb{C}$:

$$\begin{cases} iz + (1+i)w = 3 - i, \\ 2iz + (3+2i)w = 7 - 4i. \end{cases}$$

Please verify the solution in both equations.

WORKED SOLUTION

Subtract twice the first equation from the second. The coefficient of z becomes zero, and the coefficient of w becomes $(3+2i) - 2(1+i) = 1$. Thus

$$w = (7 - 4i) - 2(3 - i) = 1 - 2i.$$

Substitute into the first equation and calculate the product:

$$(1+i)(1-2i) = 1 - 2i + i - 2i^2 = 3 - i.$$

Hence $iz = (3 - i) - (3 - i) = 0$. Since $i \neq 0$, we get $z = 0$. Checking the second equation gives

$$(3+2i)(1-2i) = 3 - 6i + 2i - 4i^2 = 7 - 4i.$$

Thus $(z, w) = (0, 1 - 2i)$ solves both equations, and the elimination steps show that it is unique.

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3.2 Solving linear systems

DEFINITION 3.7

After row reduction, a variable whose column in the coefficient matrix contains a leading entry is called a **pivot variable** (or dependent variable). A variable whose column contains no leading entry is called a **free variable**. In a consistent system, every free variable can be assigned an arbitrary value in the field, after which the pivot variables are determined by the equations. Over an infinite field, a free parameter gives infinitely many choices; over a finite field, the number of choices is finite. The number of free variables is the number of degrees of freedom of a consistent system.

DEFINITION 3.8

The rank of a matrix A, denoted by $\text{rank}(A)$, is the number of nonzero rows in a row echelon form. Equivalently, it is the number of leading entries (pivots) in that form.

Remark: if a system with coefficient matrix A and n variables is consistent, its number of degrees of freedom is $n - \text{rank}(A)$. Over $\mathbb{F}_q$, it has exactly $q^{n-\text{rank}(A)}$ solutions.

EXERCISE 3.4

Please solve the following system over $\mathbb{R}$: $\begin{cases} x + 2y - 3z = 6 \\ 2x - y + 4z = 2 \\ 4x + 3y - 2z = 14 \end{cases}$.

WORKED SOLUTION

Write the system in matrix form and perform row reduction:

$$\left(\begin{array}{ccc|c} 1 & 2 & -3 & 6 \\ 2 & -1 & 4 & 2 \\ 4 & 3 & -2 & 14 \end{array}\right)$$

$$\xrightarrow[R_3 \longrightarrow R_3 - 4R_1]{R_2 \longrightarrow R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 2 & -3 & 6 \\ 0 & -5 & 10 & -10 \\ 0 & -5 & 10 & -10 \end{array}\right)$$

$$\xrightarrow{R_3 \longrightarrow R_3 - R_2} \left(\begin{array}{ccc|c} 1 & 2 & -3 & 6 \\ 0 & -5 & 10 & -10 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$$\xrightarrow{R_2 \longrightarrow \frac{-1}{5} R_2} \left(\begin{array}{ccc|c} 1 & 2 & -3 & 6 \\ 0 & 1 & -2 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$$\xrightarrow{R_1 \longrightarrow R_1 - 2R_2} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & -2 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

The matrix is now in reduced row echelon form. The first two variable columns contain leading entries, whereas the third does not.

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EXERCISE 3.4 CONTINUED

The matrix has rank 2, and the number of degrees of freedom is 1.
Therefore x, y are dependent variables, whereas the third variable z is free and may take any real value.
Returning to the system of equations,

$$\begin{aligned} x + z &= 2 \\ y - 2z &= 2 \\ 0 &= 0 \end{aligned}$$

since x, y are dependent, express them in terms of the free variable: z :

$$\begin{aligned} x &= 2 - z \\ y &= 2 + 2z \end{aligned}$$

The solution set is therefore

$$S = \{ \begin{pmatrix} 2-s \\ 2+2s \\ s \end{pmatrix} \mid s \in \mathbb{R} \} = \left\{ \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} + \begin{pmatrix} -s \\ 2s \\ s \end{pmatrix} \mid s \in \mathbb{R} \right\} = \left\{ \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} + s \cdot \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} \mid s \in \mathbb{R} \right\}$$

.
There are **infinitely many solutions**: for every choice of a real number s, substitution determines the other two coordinates, giving a different solution.
For example, setting $s = 0, 1$ gives the solutions $\begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix}$. Substitution confirms that each satisfies the original system.

EXERCISE 3.5

Please solve the following system:

$$\begin{cases} 3x_1 + 12x_2 - 6x_3 = 4 \\ x_1 + 4x_2 - 2x_3 = 1 \end{cases}$$

.

WORKED SOLUTION

Perform row reduction until a contradiction is found:

$$\left(\begin{array}{cccc} 3 & 12 & -6 & 4 \\ 1 & 4 & -2 & 1 \end{array} \right) \xrightarrow{R_2 \leftrightarrow R_1} \left(\begin{array}{cccc} 1 & 4 & -2 & 1 \\ 3 & 12 & -6 & 4 \end{array} \right)$$

$$\xrightarrow{R_2 \to R_2 - 3 \cdot R_1} \left(\begin{array}{cccc} 1 & 4 & -2 & 1 \\ 0 & 0 & 0 & 1 \end{array} \right)$$

A contradiction row has already appeared, so there is no need to complete the reduction.
Returning to the system of equations,

$$\begin{aligned} x_1 + 4x_2 - 2x_3 &= 1 \\ 0 &= 1 \end{aligned}$$

we obtain a false statement, a contradiction, so the system has no solution.

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EXERCISE 3.5 CONTINUED

In future, returning to the equations is unnecessary: recognize a **contradiction row** directly in the matrix. Such a row has zero variable coefficients and a nonzero constant term. We can immediately conclude that there is no solution.

Geometric illustration

We have encountered three possible cases:

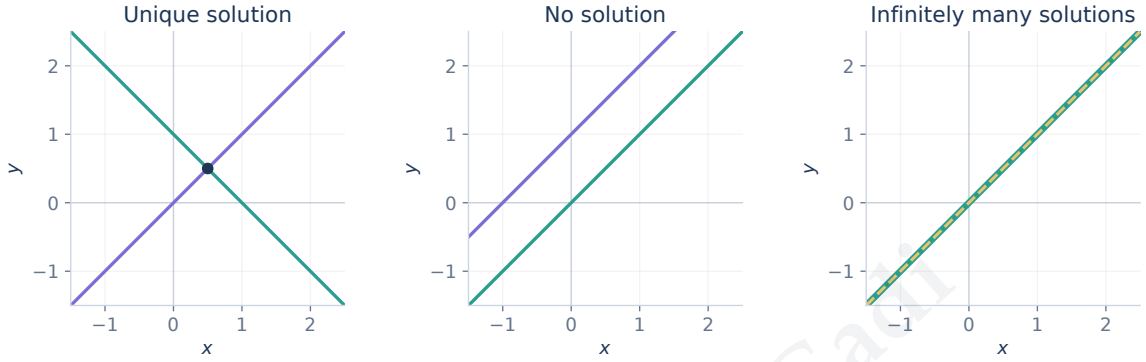


Figure 3.1. Over the real numbers, two equations in two unknowns can describe intersecting lines, distinct parallel lines, or coincident lines. These give, respectively, a unique solution, no solution, or infinitely many solutions.

EXERCISE 3.6

For which values of $a \in \mathbb{R}$ does the following system have a unique solution, infinitely many solutions, or no solution? Please state the rank of the coefficient matrix in each case.

$$\begin{cases} x - y = 0 \\ 2x + 2z = 1 \\ x + az = 1 \end{cases}$$

WORKED SOLUTION

Write the augmented matrix and perform row reduction:

$$\left(\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 2 & 0 & 2 & 1 \\ 1 & 0 & a & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - R_1} \left(\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 2 & 2 & 1 \\ 0 & 1 & a & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 1 & a & 1 \\ 0 & 2 & 2 & 1 \end{array}\right)$$

$$\xrightarrow{R_1 \to R_1 + R_2,\ ,R_3 \to R_3 - 2R_2} \left(\begin{array}{ccc|c} 1 & 0 & a & 1 \\ 0 & 1 & a & 1 \\ 0 & 0 & 2 - 2a & -1 \end{array}\right)$$

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EXERCISE 3.6 CONTINUED

We would like to make the last leading entry equal to 1, but we cannot divide by $2-2a$ before checking whether it vanishes. Distinguish the following cases:

Case 1: $2-2a=0$, that is, $a=1$. Substitution makes the third row a contradiction row, so there is no solution. (The coefficient matrix has rank 2.)

Case 2: $2-2a\neq 0$, that is, $a\neq 1$.
We can now continue the reduction by dividing by this expression to obtain a leading 1.

$$\xrightarrow{R_3\to\frac{1}{2-2a}R_3}\left(\begin{array}{ccc|c}1&0&a&1\\0&1&a&1\\0&0&1&\frac{-1}{2-2a}\end{array}\right)\xrightarrow{R_1\to R_1-aR_3,\,R_2\to R_2-aR_3}\left(\begin{array}{ccc|c}1&0&0&1+\frac{a}{2-2a}\\0&1&0&1+\frac{a}{2-2a}\\0&0&1&\frac{-1}{2-2a}\end{array}\right)$$

The solution is unique: $S=\left\{\begin{pmatrix}1+\frac{a}{2-2a}\\1+\frac{a}{2-2a}\\\frac{-1}{2-2a}\end{pmatrix}\right\}$. The matrix has rank 3 and there are no degrees of freedom.
Conclusion: for $a=1$ there is no solution; for $a\neq 1$ there is a unique solution; no value of a gives infinitely many solutions.

3.2.1 Homogeneous systems

RECALL AND TOOLS

All constant terms are 0. In this case:

- We may write only the coefficient matrix rather than the augmented matrix: the zero column is unchanged by all row operations, so there is no need to keep writing it.
- A solution always exists: no contradiction row can occur, so inconsistency is impossible. We must still determine whether the zero solution is the only solution. Over an infinite field, the existence of a nonzero solution implies infinitely many solutions.
- If there is a unique solution, it is the zero solution, also called the **trivial solution**.
- The solution set of a homogeneous system in n variables is a vector subspace of $\mathbb{F}^n$ and can always be expressed as a span.

EXERCISE 3.7

For which values of $a\in\mathbb{R}$ does the following system have a unique solution, infinitely many solutions, or no solution?

$$\begin{cases}x-y=0\\2x+2z=0\\x+az=0\end{cases}$$

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EXERCISE 3.7 CONTINUED

WORKED SOLUTION

Write only the coefficient matrix and perform row reduction. This is the same coefficient matrix as in the preceding exercise, so we immediately obtain

$$\begin{pmatrix} 1 & -1 & 0 \\ 2 & 0 & 2 \\ 1 & 0 & a \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & a \\ 0 & 1 & a \\ 0 & 0 & 2-2a \end{pmatrix}$$

As before, distinguish the following cases:

Case 1: $2-2a \neq 0$, that is, $a \neq 1$.
As before, divide by this nonzero expression to obtain a leading 1.

$$\xrightarrow{R_3 \to \frac{1}{2-2a} R_3} \begin{pmatrix} 1 & 0 & a \\ 0 & 1 & a \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_1 \to R_1 - aR_3,\, R_2 \to R_2 - aR_3} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Here the solution is unique: $S = \{ \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \}$. This is the zero solution, or trivial solution.

Case 2: $2-2a = 0$, that is, $a = 1$. Substitution gives a zero row:

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

The first two variables are dependent and the third is free. The equations are

$$\begin{cases} x+z=0 \\ y+z=0 \\ 0=0 \end{cases} \longrightarrow \begin{cases} x=-z \\ y=-z \\ z=z \end{cases} \longrightarrow S = \{ \begin{pmatrix} -s \\ -s \\ s \end{pmatrix} \mid s \in \mathbb{R} \} = \text{span} \left\{ \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} \right\}$$

Conclusion: for $a = 1$ there are infinitely many solutions; for $a \neq 1$ there is a unique, trivial solution; no value of a gives no solution.

EXERCISE 3.8 EXAM QUESTION

Work over the field $\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$, with all arithmetic modulo 5. Please solve the following system in terms of the parameter $\omega \in \mathbb{Z}_5$:

$$\begin{cases} x+y+z=1, \\ x+\omega y+z=2, \\ x+y+(\omega+1)z=1, \end{cases} \qquad x, y, z \in \mathbb{Z}_5.$$

Please determine which values of ω give a unique solution, no solution, or more than one solution. Please write the solution set explicitly in every case.

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EXERCISE 3.8 EXAM QUESTION CONTINUED

WORKED SOLUTION

Subtract the first equation from each of the other two equations. This gives the equivalent system over $\mathbb{Z}_5$:

$$\begin{cases} x+y+z=1, \\ (\omega-1)y=1, \\ \omega z=0. \end{cases}$$

The subtraction is useful because the second and third equations now each contain only one unknown. Separate the cases before dividing by a parameter-dependent expression.

1. If $\omega=1$, the second equation is $0\cdot y=1$. This is a contradiction in the field, so
$$S_1=\varnothing.$$

2. If $\omega=0$, the second equation is $-y=1$, giving $y=4$. The third equation places no restriction on z. Put $z=t\in\mathbb{Z}_5$. Then
$$x=1-y-z=1-4-t=2-t \pmod 5.$$
There are exactly five solutions:
$$\begin{aligned} S_0 &= \{(2-t,4,t) : t\in\mathbb{Z}_5\} \\ &= \{(2,4,0),(1,4,1),(0,4,2),(4,4,3),(3,4,4)\}. \end{aligned}$$

3. If $\omega\in\{2,3,4\}$, both ω and $\omega-1$ are nonzero, so both are invertible in the field. We obtain
$$z=0, \qquad y=(\omega-1)^{-1}, \qquad x=1-(\omega-1)^{-1}.$$
Calculate the inverses explicitly:
$$1^{-1}=1, \qquad 2^{-1}=3, \qquad 3^{-1}=2 \pmod 5.$$
Thus each of these three cases has a unique solution:
$$S_2=\{(0,1,0)\}, \qquad S_3=\{(3,3,0)\}, \qquad S_4=\{(4,2,0)\}.$$

As an additional check, the determinant of the coefficient matrix is $\omega(\omega-1)$. Only $\omega=0,1$ therefore require a distinction between inconsistency and multiple solutions.

3.3 Vector spaces

RECALL AND TOOLS

1. To define a vector space V over a field $\mathbb{F}$, we specify an addition of vectors in V and a multiplication of a vector in V by a scalar in $\mathbb{F}$, satisfying the vector space axioms. These are **in addition** to addition and multiplication in $\mathbb{F}$. A product of two vectors is not part of the vector space axioms; if needed later, it will be defined separately.

2. Every field is a vector space over itself, with the field's addition and multiplication.

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3. A vector space V is always defined over a field $\mathbb{F}$. Always specify the field of scalars. For example, as discussed in the lecture, $\mathbb{C}$ can be regarded as a vector space over $\mathbb{C}$ or over $\mathbb{R}$. These are two different vector spaces.

EXAMPLE 3.3

Let $\mathbb{F}$ be a field.

(a) $\mathbb{F}^n$: the set of n-tuples over $\mathbb{F}$, where $n \in \mathbb{N}$.

(b) $M_{m\times n}(\mathbb{F})$: the set of $m \times n$ matrices over $\mathbb{F}$.

(c) $\mathbb{F}[X]$: the set of all polynomials with coefficients in $\mathbb{F}$.

EXERCISE 3.9

Please determine whether the following sets are subspaces of $\mathbb{R}[X]$ over $\mathbb{R}$, with the usual operations:

1. $V = \{p(X) \in \mathbb{R}[X] : p(5) = 0, p(4) \neq 0\}$.
2. $V = \{p(X) \in \mathbb{R}[X] : \exists c \in \mathbb{R}, p(c) = 0\}$.

WORKED SOLUTION

1. No. The zero polynomial is absent, since its value at 4 is 0. Addition also fails to be closed: the polynomials $p(X) = (X-5)(X-3)$ and $q(X) = (X-5)^2$ belong to the set, because $p(4) = -1$ and $q(4) = 1$, but
$$(p+q)(X) = (X-5)(2X-8), \qquad (p+q)(4) = 0.$$
therefore $p+q$ does not belong to the set.
2. No. The set is not closed under addition. For example, $(X+1)^2, -2X \in V$, but $(X+1)^2 - 2X = X^2 + 1 \notin V$.

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3.4 Subspaces

DEFINITION 3.9

Let V be a vector space over a field $\mathbb{F}$, and let $U \subseteq V$ be a subset that is itself a vector space over $\mathbb{F}$ under the addition and scalar multiplication inherited from V. Then U is a vector subspace of V.

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PROPOSITION 3.2

Let V be a vector space over a field $\mathbb{F}$, and let $U \subseteq V$ be a subset. Then U is a vector subspace if and only if all of the following conditions hold:

1. $0 \in U$, where this is the zero vector of V.
2. U is closed under addition.
3. U is closed under scalar multiplication.

EXAMPLE 3.4

1. $V = \{p(x) \in \mathbb{F}[X] | p(5) = 0\}$ is a subspace of $\mathbb{F}[X]$.
   Proof:
   (a) Clearly $0 \in V$.
   (b) Suppose $p(x), q(x) \in V$. Define $r(x) = p(x) + q(x)$. Then $r(5) = p(5) + q(5) = 0$, so $r(x) \in V$.
   (c) Suppose $p \in V$ and let $\alpha \in \mathbb{F}$. Define $r = \alpha p$. Then $r(5) = \alpha p(5) = 0$.
2. $V = \{(x_1, \ldots, x_n) \in \mathbb{R}^n | x_2 = 0\}$ (where $n \geq 2$). This is a subspace of $\mathbb{R}^n$.
   Proof:
   (a) Clearly $0 \in V$.
   (b) Let $x = (x_1, \ldots, x_n), y = (y_1, \ldots, y_n) \in V$. Then $x_2 = y_2 = 0$, so the second coordinate of $x + y$ is 0; hence $x + y \in V$
   (c) If $x \in V$ and $\alpha \in \mathbb{R}$, then $\alpha x_2 = 0$, so $\alpha x \in V$.

EXERCISE 3.10

Please prove or disprove:

1. The set $U = \left\{ \begin{pmatrix} a \\ a^2 \end{pmatrix} \mid a \in \mathbb{R} \right\}$ is a vector subspace of $\mathbb{R}^2$.
2. The set of symmetric matrices $Sym_n(\mathbb{F}) := \left\{A \in M_{n\times n}(\mathbb{F}) \mid A = A^T\right\}$ is a vector subspace of $M_{n\times n}(\mathbb{F})$.

WORKED SOLUTION

1. This is not a vector subspace, because addition is not closed. For example, $\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1^2 \end{pmatrix} \in U$, but
$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} \notin U$$
2. The zero matrix is symmetric. Suppose $A, B \in \text{Sym}_n(\mathbb{F})$ and $\lambda \in \mathbb{F}$. Then
$$(A + B)^T = A^T + B^T = A + B,$$

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EXERCISE 3.10 CONTINUED

so $A + B$ is symmetric. Also,

$$(\lambda A)^T = \lambda A^T = \lambda A,$$

so λA is symmetric. All three conditions of the subspace test hold.

EXERCISE 3.11

An engineering sensing system models a normalized signal by a real polynomial of degree at most 3:

$$p(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3.$$

Time t is measured in normalized time units. This is a mathematical approximation of a signal, without assuming any particular sensor type.

1. Please find all polynomials satisfying

$$p(0) = 0, \quad p(1) = 0.2, \quad p'(0) = 0, \quad p'(0.5) = 0.$$

Is the solution set a subspace of $\mathbb{R}[t]$? Is $t = 0.5$ a local maximum or minimum point?

2. Repeat the question when the measurements change to

$$p(0) = 0.1, \quad p(1) = 0.3, \quad p'(0) = 0, \quad p'(0.5) = 0.$$

3. In the second part, add the measurement $p'(1) = 0.4$. Is the system still consistent? Is its solution set a subspace?

WORKED SOLUTION

Differentiate the polynomial:

$$p'(t) = a_1 + 2a_2 t + 3a_3 t^2.$$

Part 1. The conditions at $t = 0$ give $a_0 = a_1 = 0$. The remaining two conditions give

$$\begin{cases} a_2 + a_3 = 0.2, \\ a_2 + 0.75 a_3 = 0. \end{cases}$$

Subtract the second equation from the first:

$$0.25 a_3 = 0.2 \quad \Longrightarrow \quad a_3 = 0.8, \qquad a_2 = -0.75 \cdot 0.8 = -0.6.$$

Thus there is a unique solution,

$$p_1(t) = -0.6 t^2 + 0.8 t^3.$$

Check: $p_1(1) = -0.6 + 0.8 = 0.2$ and $p_1'(0.5) = -1.2 \cdot 0.5 + 2.4 \cdot 0.25 = 0$. The solution set is not a subspace: the zero polynomial does not satisfy $p(1) = 0.2$. The condition $p'(0.5) = 0$ alone gives a stationary point, which need not be a peak. Here,

$$p_1''(t) = -1.2 + 4.8t, \qquad p_1''(0.5) = 1.2 > 0,$$

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EXERCISE 3.11 CONTINUED

so this is a local minimum.

Part 2. This time $a_0 = 0.1$ and $a_1 = 0$. The condition $p(1) = 0.3$ gives $a_2 + a_3 = 0.2$, so the two equations for a_2, a_3 are identical to those in the first part. We obtain

$$p_2(t) = 0.1 - 0.6t^2 + 0.8t^3 = p_1(t) + 0.1.$$

Again, the solution is unique, and its solution set is not a subspace because it does not contain the zero polynomial. The derivatives of p_1, p_2 agree, so $t = 0.5$ is again a local minimum point.

Part 3. The unique polynomial from part 2 satisfies

$$p_2'(1) = -1.2 + 2.4 = 1.2 \neq 0.4.$$

Thus the additional condition introduces a contradiction. There is no solution, and the empty solution set is not a subspace because it does not contain the zero vector. By contrast, $\{0\}$ is a subspace.

3.5 Supplementary exercises: measurements and counting solutions

EXERCISE 3.12

Three sensors measure sums of real state variables:

$$y_1 = x_1 + x_2 = 5, \qquad y_2 = x_2 + x_3 = 7, \qquad y_3 = x_1 + x_3 = 6.$$

Please recover the three state variables. A control sensor measures $y_4 = x_1 + x_2 + x_3$. What is its consistent reading, and what does a reading $y_4 = 10$ imply about the exact model?

WORKED SOLUTION

Add the first two equations and subtract the third:

$$2x_2 = 5 + 7 - 6 = 6, \qquad x_2 = 3.$$

Hence $x_1 = 5 - 3 = 2$ and $x_3 = 7 - 3 = 4$. The third equation gives $2 + 4 = 6$. The sum of the three readings is twice the sum of the state variables:

$$y_1 + y_2 + y_3 = 2(x_1 + x_2 + x_3) = 2y_4.$$

The consistent reading is therefore $y_4 = (5 + 7 + 6)/2 = 9$. The reading 10 contradicts this, so the four equations are inconsistent in the exact model. This consistency check can indicate a measurement error or an unsuitable model.

EXERCISE 3.13

Please solve the following system over $\mathbb{Z}_2$:

$$\begin{cases} x + y + z = 0, \\ x + z = 1. \end{cases}$$

How many solutions are there? Please compare with the system given by the same equations over $\mathbb{R}$.

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EXERCISE 3.13 CONTINUED

WORKED SOLUTION

Over $\mathbb{Z}_2$, subtracting the second equation from the first gives $y = -1 = 1$. Put $z = t$; then $x = 1 - t = 1 + t$. Thus

$$(x, y, z) = (1 + t, 1, t), \qquad t \in \mathbb{Z}_2.$$

There are exactly two solutions: $(1, 1, 0)$ and $(0, 1, 1)$. Over $\mathbb{R}$, subtraction gives $y = -1$, and we obtain

$$(x, y, z) = (1 - t, -1, t), \qquad t \in \mathbb{R}.$$

Here there are infinitely many solutions. Both systems have one free variable; the number of solutions depends on the field.

Additional practice

EXERCISE 3.14 FOUNDATION

Please solve the following system over $\mathbb{R}$:

$$\begin{cases} x + y + z = 4, \\ x - y + z = 2. \end{cases}$$

Please write the solution set as a particular solution plus a multiple of a fixed vector. Please determine whether it is a subspace, and please describe the set of differences of two solutions.

WORKED SOLUTION

Subtracting the second equation from the first gives $2y = 2$, so $y = 1$. Substitution into the second equation gives $x + z = 3$. Put $z = t \in \mathbb{R}$. Then

$$S = \{(3 - t, 1, t) : t \in \mathbb{R}\} = (3, 1, 0) + \text{span}\{(-1, 0, 1)\}.$$

These are all solutions because each subtraction and substitution is equivalent to the original system. The set is not a subspace: the zero vector does not satisfy the equations. The difference of solutions with parameters t, s is

$$(3 - t, 1, t) - (3 - s, 1, s) = (t - s)(-1, 0, 1).$$

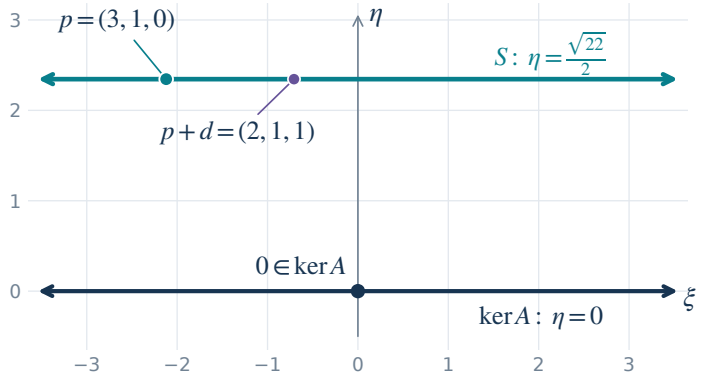
Every real multiple occurs, for example by setting $s = 0$. Thus the difference set is exactly $\text{span}\{(-1, 0, 1)\}$, which is the solution set of the associated homogeneous system.

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$$S=\{(3-t,1,t):t\in\mathbb{R}\}=p+\ker A$$

$$\Phi(\xi,\eta)=\xi u+\eta v,\qquad u=\frac{(-1,0,1)}{\sqrt{2}},\quad v=\frac{(3,2,3)}{\sqrt{22}}$$



$$\ker A=\operatorname{span}\{d\},\quad d=(-1,0,1),\qquad 0\notin S$$

Figure 3.2. The real system $x+y+z=4$, $x-y+z=2$ has solution set $S=p+\ker A$, where $p=(3,1,0)$ and $\ker A=\operatorname{span}\{(-1,0,1)\}$. The ξ,η axes describe the plane $\operatorname{span}\{u,v\}\subseteq\mathbb{R}^3$ through the map Φ displayed in the figure. The vectors u,v are perpendicular unit vectors, so the chart preserves lengths and angles. The translated line excludes the origin, whereas the kernel contains it.

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EXERCISE 3.15 CHALLENGE

Let $a,b\in\mathbb{R}$. Please classify the solutions of

$$\begin{cases}x+y+z=2,\\ x+ay+z=b,\\ x+y+az=b.\end{cases}$$

In each case, please find the coefficient matrix rank, the number of free variables when consistent, and the solution set. Also find the kernel of the coefficient matrix, and please determine whether the solution set is a subspace.

WORKED SOLUTION

Subtract the first equation from both the second and the third. The equivalent system is

$$\begin{cases}x+y+z=2,\\ (a-1)y=b-2,\\ (a-1)z=b-2.\end{cases}$$

1. If $a\neq 1$, division by $a-1$ is permitted. Put $c=(b-2)/(a-1)$. The unique solution is

   $$S=\{(2-2c,c,c)\}.$$

   There are three pivots, so the rank is 3 and there are no free variables. In the homogeneous system, $y=z=0$ and then $x=0$, so $\ker A=\{0\}$.

2. If $a=1$ and $b=2$, the last two equations after subtraction are $0=0$. The rank is 1, and

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EXERCISE 3.15 CHALLENGE CONTINUED

there are two free variables. Choosing $y = s, z = t$ gives

$$S = \{(2 - s - t, s, t) : s, t \in \mathbb{R}\} = (2, 0, 0) + \text{span}\{(-1, 1, 0), (-1, 0, 1)\}.$$

The kernel is $\text{span}\{(-1, 1, 0), (-1, 0, 1)\}$.

3. If $a = 1$ and $b \neq 2$, the equation $0 = b - 2$ is a contradiction, so $S = \varnothing$. The coefficient matrix still has rank 1, while the augmented matrix has rank 2. The kernel depends only on A, so it is the same kernel as in the preceding case.

The solution set is never a subspace: in the consistent cases the zero vector fails to satisfy $x + y + z = 2$, and in the inconsistent case the set is empty. A number of degrees of freedom of a solution set cannot be defined through a choice of solution parameters when no solutions exist.

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EXERCISE 3.16 CHALLENGE

Let $S \subseteq \mathbb{R}^n$ be nonempty and assume $(1-t)x + ty \in S$ for all $x, y \in S$ and $t \in \mathbb{R}$.

1. Please choose $a \in S$ and prove that $U = \{x - a : x \in S\}$ is a subspace and $S = a + U$, where $a + U = \{a + u : u \in U\}$.
2. Is it enough to require only $(x + y)/2 \in S$ for all $x, y \in S$? Please prove or disprove.

WORKED SOLUTION

We have $0 = a - a \in U$. For $u \in U$ and $\lambda \in \mathbb{R}$, the hypothesis gives

$$(1 - \lambda)a + \lambda(a + u) = a + \lambda u \in S,$$

so $\lambda u \in U$. For $u, v \in U$, taking the midpoint of $a + u$ and $a + v$ gives $(u + v)/2 \in U$. Scalar closure with scalar 2 then gives $u + v \in U$. Hence U is a subspace. Its definition gives both inclusions in $S = a + U$. Midpoint closure alone is insufficient. The set $\mathbb{Q} \subseteq \mathbb{R}$ is closed under midpoints. If it were $a + U$ for a real subspace U, the fact that $0 \in \mathbb{Q}$ would imply $\mathbb{Q} = U$. But $1 \in U$ would then imply $\sqrt{2} \cdot 1 \in U$, a contradiction.

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EXERCISE 3.17 CHALLENGE

Consider a finite linear system over a field F with $n \geq 1$ unknowns. Assume every collection of at most $n+1$ of its equations is consistent. Please prove that the full system is consistent. Please construct an inconsistent system of $n+1$ equations for which every collection of n equations is consistent.

WORKED SOLUTION

Suppose the system is inconsistent and choose an inconsistent collection of minimum size k. Write its equations as

$$\sum_{i=1}^{n} a_{ji} x_i = b_j, \qquad 1 \leq j \leq k.$$

The hypothesis gives $k > n+1$. Consider the homogeneous system in $\lambda_1, \ldots, \lambda_k$ given by

$$\sum_{j=1}^{k} \lambda_j a_{ji} = 0 \quad (1 \leq i \leq n), \qquad \sum_{j=1}^{k} \lambda_j b_j = 0.$$

It has $n+1$ equations and more than $n+1$ unknowns. Row reduction therefore gives a nonzero solution. Choose j with $\lambda_j \neq 0$. These identities express equation j, including its right-hand side, as a linear combination of the other equations. Hence every solution of the other equations also satisfies equation j. Minimality of k says that the other equations have a solution. That solution satisfies the whole collection, a contradiction.
The bound is sharp. Take

$$x_1 = 0, \ldots, x_n = 0, \qquad x_1 + \cdots + x_n = 1.$$

The full system is inconsistent. If the final equation is omitted, zero is a solution. If $x_j = 0$ is omitted, setting $x_j = 1$ and all other unknowns to zero gives a solution. Thus every collection of n equations is consistent.

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